Lesson for the course “Human-Computer Interaction”

Discovering requirements

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Purpose of the product's requirements activity

- What is the purpose of the requirements activity?

1. **Explore the problem space**
   - Recall the PACT framework

2. **Establish a description of what will be developed**

How to discover product's requirements

Data gathering



Data analysis



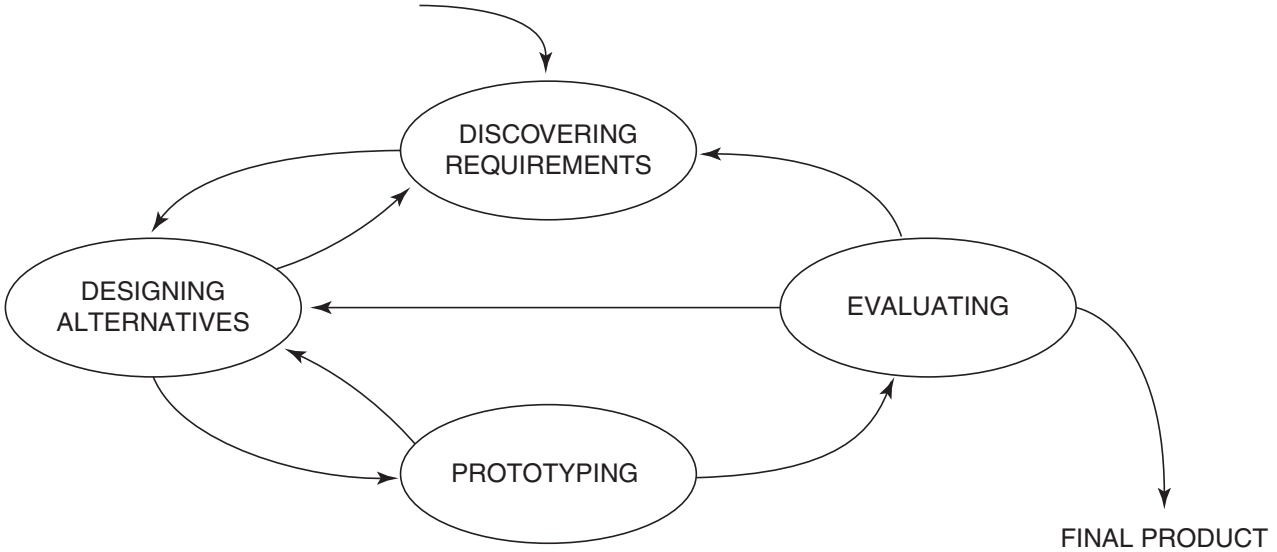
Results



Expression of results as requirements

The product's requirements

- In the interaction design lifecycle, requirements discovery is **iterative**
  - The iterative cycles ensure that the **lessons learned** from any of these activities feed into each other
  - In practice, requirements **evolve** and develop as the stakeholders interact with designs and learn what is possible and how features can be used



The importance of requirements

- Whenever there is a **team** working on a project, it is important for everybody to have a **shared understanding** of what people are working on
- A list of user requirements is useful to have a **common ground** and documentation
- To **avoid the high cost of fixing errors** late in the process, i.e., at the moment of software development

Common mistakes in the requirements discovery phase

- **The underestimation of complexity**

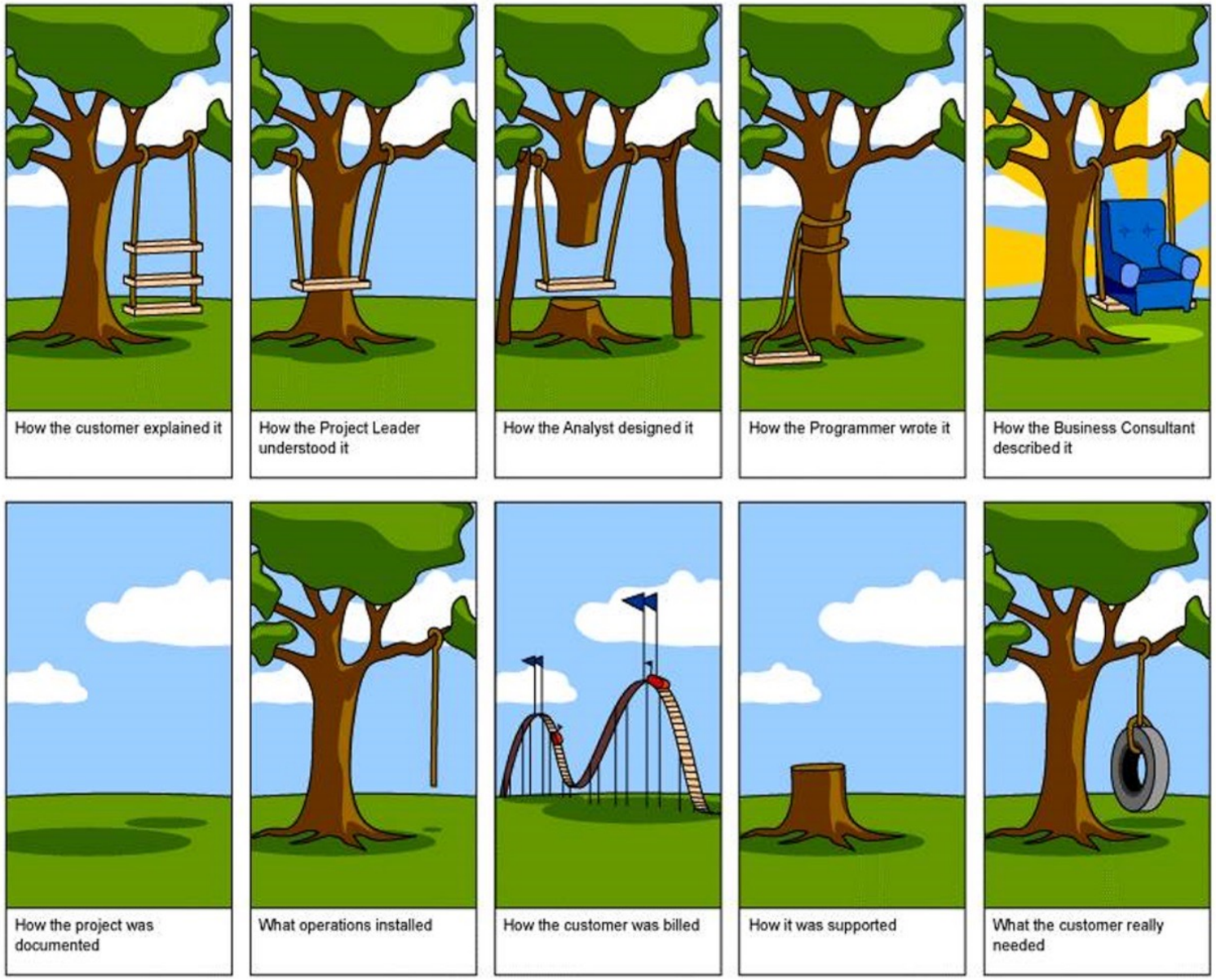




- Many crowdfunding project have failed…

Common mistakes in the requirements discovery phase

- **Failure to engage with stakeholders**
  - Insufficient users' communication:
    - stakeholders need to provide key input to the critical decisions made in the project – so they need to be involved and to be asked the right questions
  - Insufficient analysis

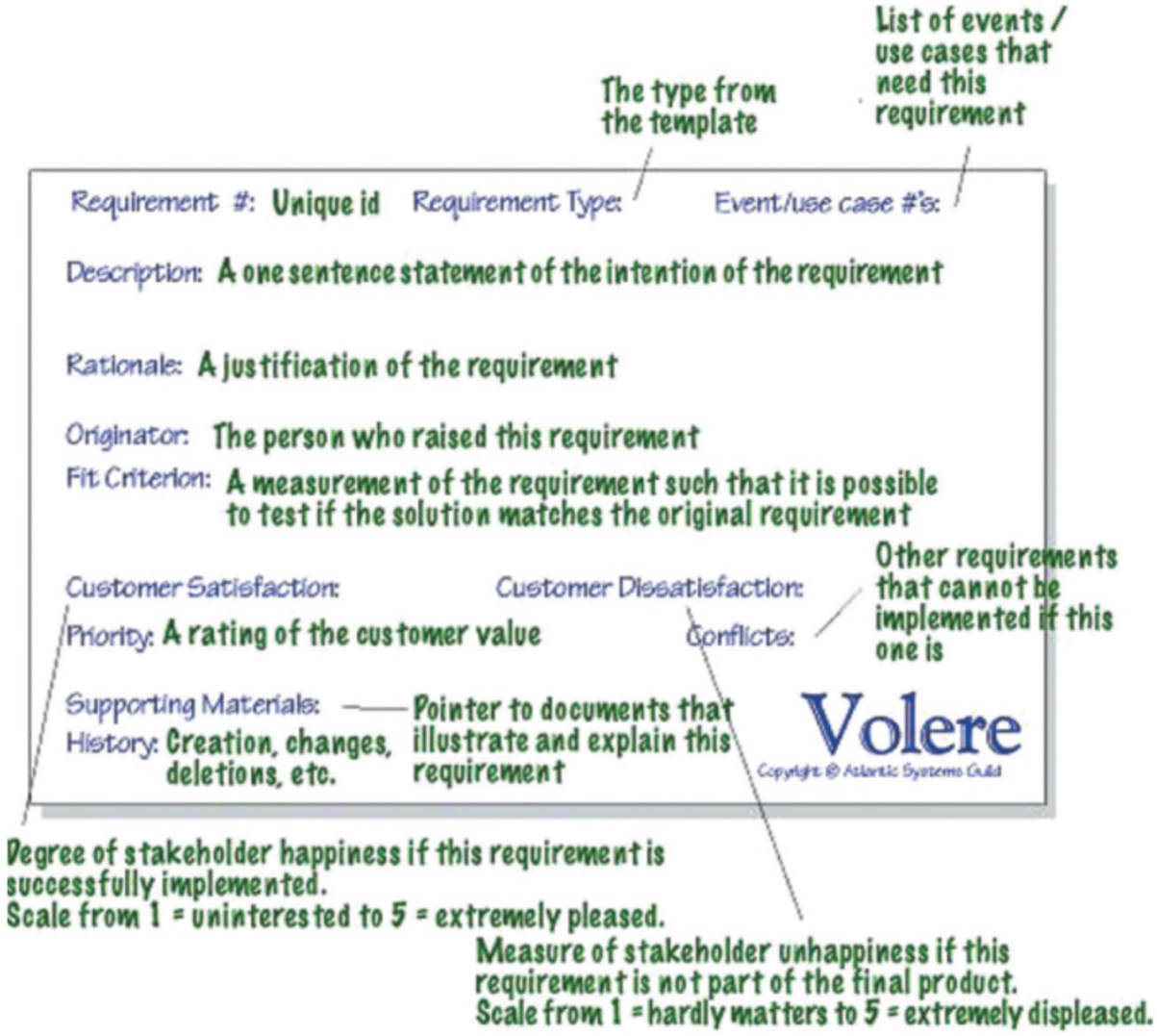


- Requirements activity is the stage where **mis-communication** occurs most commonly

What are requirements?

- A requirement is a **statement** about an intended product that specifies what it is expected to do or how it will perform
- Requirements should be as specific, unambiguous, and clear as possible
- Different forms and different levels of abstraction
  - **Atomic requirements shell**
    - https://www.volere.org/
  - **User stories**
- Different kinds of requirements
  - **Functional requirements**
  - **Non functional requirements**

The structure of an atomic requirements shell



Volere Shell

Requirement #: **75** Requirement Type: **9** Event/use case #: **6**

Description: **The product shall issue an alert if a weather station fails to transmit readings.**

Rationale: **Failure to transmit readings might indicate that the weather station is faulty and needs maintenance, and that the data used to predict freezing roads may be incomplete.**

Source: **Road Engineers**
Fit Criterion: **For each weather station the product shall communicate to the user when the recorded number of each type of reading per hour is not within the manufacturer's specified range of the expected number of readings per hour.**

Customer Satisfaction: **3** Customer Dissatisfaction: **5**

Dependencies: **None** Conflicts: **None**

Supporting Materials: **Specification of Rosa Weather Station**

History: **Raised by GBS, 28 July**

Volere

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User stories

- User stories are short, simple descriptions of a feature told from the perspective of the person who desires the new capability (usually a user or customer of the system)
- Format:

  As a <role>, I want <behaviour> so that <benefit>

- Example of user stories for a travel organizer might be:
  - *As a <traveler>, I want <to save my favourite airline for all my flights> so that <I will be able to collect air miles>*
  - *As a <travel agent>, I want <my special discount rates to be displayed to me> so that <I can offer my clients competitive rates>*

User stories

- A user story serves as a **starting point for a conversation** to extend and clarify requirements
- User stories may also be used to capture usability and user experience goals
- User stories are most prevalent in **agile development** contexts
  - A user story forms the basis of planning for a sprint and contains just enough information so that the developers can produce a reasonable estimate of the effort to implement it

Epic user stories

- During the early stages of the IxD process, requirements may emerge in the form of "epics"
- An epic is a user story that may take weeks or months to implement
- **Epics will be broken down** into smaller chunks of effort (user stories), before they are pulled into a sprint
- Examples of epics for a travel organizer app might be the following:
  - *As a <group traveler>, I want <to choose from a range of potential vacations that suit the group's preferences> so that <the whole group can have a good time>*
  - *As a <group traveler>, I want <to know the visa restrictions for everyone in the group> so that <visas can be arranged for everyone in the group in plenty of time>*

Different kinds of requirements

- **Functional requirements**
  - Describe what the system should do

- **Non-functional requirements**
  - Describe the characteristics of the product:
    - Data requirements
    - Environment requirements (or context of use)
    - User characteristics
    - Usability goals
    - User experience goals

Non-functional requirements

- **Data requirements**
  - They capture the type, volatility, size/amount, persistence, accuracy, and value of the required data (all interactive products have to handle some data)
  - What kinds of data need to be stored?
  - How will they be stored (e.g., database)?

Non-functional requirements

- **Environment requirements (or context of use)**
  - They refer to the circumstances in which the product will operate
  - **Physical environment**: dusty? noisy? vibration? light? heat? humidity? …. (e.g., in a hospital)
  - **Social environment**: collaboration and co-ordination, data sharing, distributed, synchronous or asynchronous, privacy
  - **Organizational environment**: user support, communications structure and infrastructure, availability of training
  - **Technical environment**: what technologies will it run on or need to be compatible with?

Non-functional requirements

- **Users Characteristics**
- What are the characteristics of the users?
  - Characteristics: nationality, educational background, attitude to computers
  - Utilization of the system
    - **Novices** (e.g., may prefer step-by-step guidance)
    - **Experts** (e.g., may prefer a flexible interaction with more wide-ranging powers of control)
    - **Frequent** (e.g., may want to use short cuts)
    - **Casual/infrequent** (e.g., may prefer clear menu paths)

Non-functional requirements

- **Usability goals**
  - Effective to use (**effectiveness**)
  - Efficient to use (**efficiency**)
  - Safe to use (**safety**)
  - Having good utility (**utility**)
  - Easy to learn (**learnability**)
  - Easy to remember how to use (**memorability**)
- **User experience goals**
  - Satisfying
  - Fun
  - Enjoyable
  - Entertaining
  - Helpful
  - Motivating
  - Aesthetically pleasing
  - Enhancing sociability
  - Rewarding
  - Support creativity
  - Emotionally fulfilling
- Note: these are just general remarks: you need to specify these requirements, give details for your system

Tips for the exam

- List (as bullet points) the functional and non functional requirements you derived from the data gathering process of the discovery requirement phase, E.g.
  - Functional requirements:
    - *"The system must allow the users to login via fingerprints"*
    - *…*
  - *Non functional requirements:*
    - *"The system must be engaging for both experts and novices"*
    - *….*
- However, you are encouraged to use the Volere shell template and/or the user stories, and provide a link to them on the Google Drive folder (you don't have space to fit everything in the report, 12 pages maximum)

Bringing requirements to life

- Goal: to augment the basic requirements expressed as user stories, in Volere template, or in other form
- **Personas**
  - Rich descriptions of typical users, not specific people
- **Scenarios**
  - An informal narrative story, simple, 'natural', personal, not generalizable
- Typically personas and scenarios are used together
- They complement each other in order to bring realistic detail that allow the designer to explore the user's current activities, future use of new products, and futuristic visions of new technologies

Family traveler



""I want a travel organiser that will offer me a range of potential vacations that suit our needs"

Age: 35
Work: Plumber
Family: Married, two children

Personality

Introvert — Extrovert
Thinking — Feeling
Sensing — Intuition

Organised | Practical | Expects high standard

Goals

- To book comprehensive travel quickly
- To find a trip that meets the needs of the whole family
- To feel supported and guided from the beginning of the booking experience right to the end.

Frustrations

- Wasting time filling in forms
- Too much irrelevant information
- Existing systems tend to be too diverse and complicated

Bio

Will loves to take his family on adventure holidays to explore new challenges. His children, Sky (8) and Eamonn (15) are old enough to take part in several sporting activities and he wants to make the most of this before they no longer want to go on trips with him and his wife, Claire. He likes the fact that choosing travel options is so much easier than it used to be, but is frustrated by the many different sources and disjointed options that this can result in. He wants a travel organiser that can provide clear support for family holidays while offering as wide a choice as possible.

Motivation

Price
Comfort
Choice

Favourite destinations





Personas

- They are rich descriptions of typical users, which capture a set of user characteristics and needs
  - "Typical", **not idealized**
- They are synthesized **from real people,** based on user research
- Bring personas to life with
  - **Name**
  - **Characteristics** (e.g., age, attitude, motivations, key needs)
  - **Goals**
  - **Personal background** (e.g., level of familiarity with technology)
  - **Picture**
- The kind of information they include needs to be pertinent to the product under development
- Good personas help designer with design decisions and remind the team about who will use the product

Exercise (10 minutes)

- Have a look to the personas illustrated in the next few slides
- Reflect on each field, so you can take inspiration on how to create your personas

- Note: you will have to create a small set of personas for the exam, for
  - **Primary users**(represent a large section of the intended user group)
  - Possibly also for
    - **Secondary users**
    - **Tertiary users**

CAPLIN

Ginnie

BACKGROUND

- 15, Female
- Ongoing Private Education
- Ambitious
- Comfortable using technology to communicate

MOTIVATIONS

- Keeping in touch with her network
- Fashion/street cred
- Keeping up with peers.

FRUSTRATIONS

- Sad people trying to be 'friends' on Facebook
- Having to be in bed @ 11pm
- Being swamped in friends updates
- Missing important status updates



Receives private tutoring in Maths and English as these are not her strong subjects. Enjoys playing for the school's 2nd teams for netball and Lacrosse and is good at art.

She loves recording her favourite shows: ER and Sun Valley High on Sky+ and spends some of her time on her Laptop that Daddy bought her watching videos on YouTube, downloading music, keeping up to date with her friends on Facebook and chatting via MS IM to her cousin who is at University in Leeds.

She loves Ugg boots and Abercrombie & Fitch and uses the Internet to shop and find the cheapest prices.

"I want to easily hook up with my friends whilst watching TV"



















CARLO (PLANNER)

I'm going to spend some days in an accessible thermal area, where I can do some particular sport activities and follow the special diet I need.

ATTITUDE AND MOTIVATIONS

He's a very energetic hard-working man. He's very health-conscious and practises many treatments which help him with his motor disability.

DIS/ABILITIES

He is on a wheelchair. However, this fact doesn't prevent him from doing almost all the activities he loves.

TECHNOLOGY PROFICIENCY

He has got an iPhone and a Mac. He's very good in technology because of his work.

RESOURCES

He doesn't have a car nor a driving licence.

PERSONAL CONTEXT

He travels with his wife.

KEY-NEEDS

I have to identify places (such as restaurants, hotels, etc.) and activities that have facilities for disabled people;

I have to monitor different parameters related to sport activities (i.e. burned calories) and my diet.







FRANCESCO / EXPLORER

I'm preparing a new digital guide about the best restaurants in Trentino, with a specific focus on small and unknown places.

ATTITUDE AND MOTIVATIONS

He's fanatic of good food and very curious.

DIS/ABILITIES

He's a chef, but he doesn't know very well mountain food traditions.

TECHNOLOGY PROFICIENCY

He has a smartphone and a tablet and is quite experienced in the use of several applications to collect information..

RESOURCES

He has a lot of money, his personal car and a GPS.

PERSONAL CONTEXT

He travels alone.

KEY-NEEDS

I have to discover new places not already mentioned in mainstream guides;
I have to access to people reviews and comments;
I have to collect and store different materials (comments, interviews, pictures, maps, etc.) related to the restaurants I've tried;
I have to meet local expert or local community to discuss about local food traditions and keep track of this interations;

Scenarios

- A scenario is an **informal narrative description**
- It describes human activities or tasks **in a story** that allows for the exploration and discussion of contexts, needs, and requirements
- It does not necessarily describe the use of software or other technological support used to achieve a goal
- Using **the vocabulary and phrasing of users** means that scenarios can be understood by stakeholders, and they are able to participate fully in development

Scenarios

- Telling stories is a natural way for people to explain what they are doing

- The focus of such stories is also naturally likely to be about what the users are trying to achieve, that is, their **goals**

- **Understanding** why people do things as they do and what they are trying to achieve in the process, **focuses the study on human activity rather than interaction with technology**

Scenario for group travel organizer

"The Thomson family enjoy outdoor activities and want to try their hand at sailing this year. There are four family members: Sky (8 years old), Eamonn (12 years old), Claire (32), and Will (35). One evening after dinner they decide to start exploring the possibilities. They want to discuss the options together but Claire has to visit her elderly mother so will be joining the conversation from her mother's house down the road. As a starting point, Will enters an idea they had been discussing over dinner– a sailing trip for four novices in the Mediterranean. The system supports users to log on from different locations and use different devices so that all members of the family can interact easily and comfortably with it wherever they are. The system's initial suggestion is a flotilla, where several crews (with various levels of experience) sail together on separate boats. Sky and Eamonn aren't very happy at the idea of going on vacation with a group of other people, even though the Thomsons would have their own boat. The travel organizer shows them descriptions of flotillas from other children their ages and they are all very positive, so eventually, everyone agrees to explore flotilla opportunities. Will confirms this recommendation and asks for detailed options. As it's getting late, he asks for the details to be saved so everyone can consider them tomorrow. The travel organizer emails them a summary of the different options available."

Scenarios

- May be textual descriptions, animations, audio or video

Example of scenario in a video

https://vimeo.com/135352495



Examples of animation scenarios

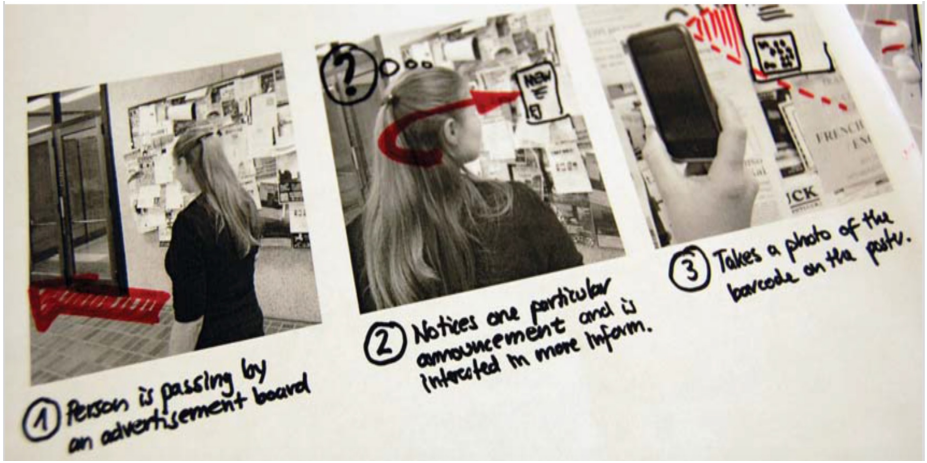


Scenarios: two types

- **Activity scenarios**
  - Describe the activity as it is at the moment we are investigating it (what users currently do to carry on their activities)
  - In case of a scenario represented through a visual medium, the pictures and videos are aimed to document it, so they are very realistic
- **Envisioning scenarios**
  - Describe how the users' activity would change by that moment that we introduce our new technology in their lives
  - Envisioning scenarios are imaginary, they represent a possible world, so sometimes when they are represented visually through pictures and videos they are displayed with filters, blurs, etc.

- *Envisioning scenario example:* two different ways to represent visually a scenario where a person can interact with an advertisement through a QR code. By taking a picture through a smartphone they receive more detailed information on that product on their smartphone





Envisioning scenario: the Knowledge Navigator by Apple (1987)

- Video: https://youtu.be/umJsITGzXd0



- In 1987 predicted several technologies that we use today

Envisioning scenario: the Knowledge Navigator by Apple (1987)

- The scenario describes a device that can access a large networked database of hypertext information, and use software agents to assist searching for information
- Involves a tablet style computer with numerous advanced capabilities, including an excellent text-to-speech system, a gesture based interface resembling the multi-touch interface and a powerful speech understanding system, allowing the user to converse naturally with the system
- In 1987 there was no Internet, no tablets, no video-calls with screen-sharing and collaborative working, no intelligent software agents, and little research on voice based interfaces

Envisioning scenario: the Knowledge Navigator by Apple (1987)

- It was deeply criticized at that time, and also several years later…
- The discussions around the Knowledge Navigator and the technology-prediction videos that followed were hated, especially in the field of human-computer interaction
- Major criticism: the Knowledge Navigator is an unrealistic portrayal of the abilities and capacities of any software agent, thus setting the expectations way too high
- Critics claimed that in the foreseeable and even distant future, technology won't be able to deliver anywhere close to these "promises". Especially the human-likeness of the interaction was said to create a misleading concept and idea of future human-computer interaction… *but as we know, the history proved them wrong*

Scenarios and personas

- **A scenario describes one use of a product or one example of achieving a goal, while a persona characterizes a typical user of the product**
- The scenarios and personas are **tightly linked**
- Each scenario represents a single experience of using the product from the perspective of one persona
  - For the exam, you can choose to do the scenarios as a graphical set of vignettes, a textual description or a video

Scenarios and personas

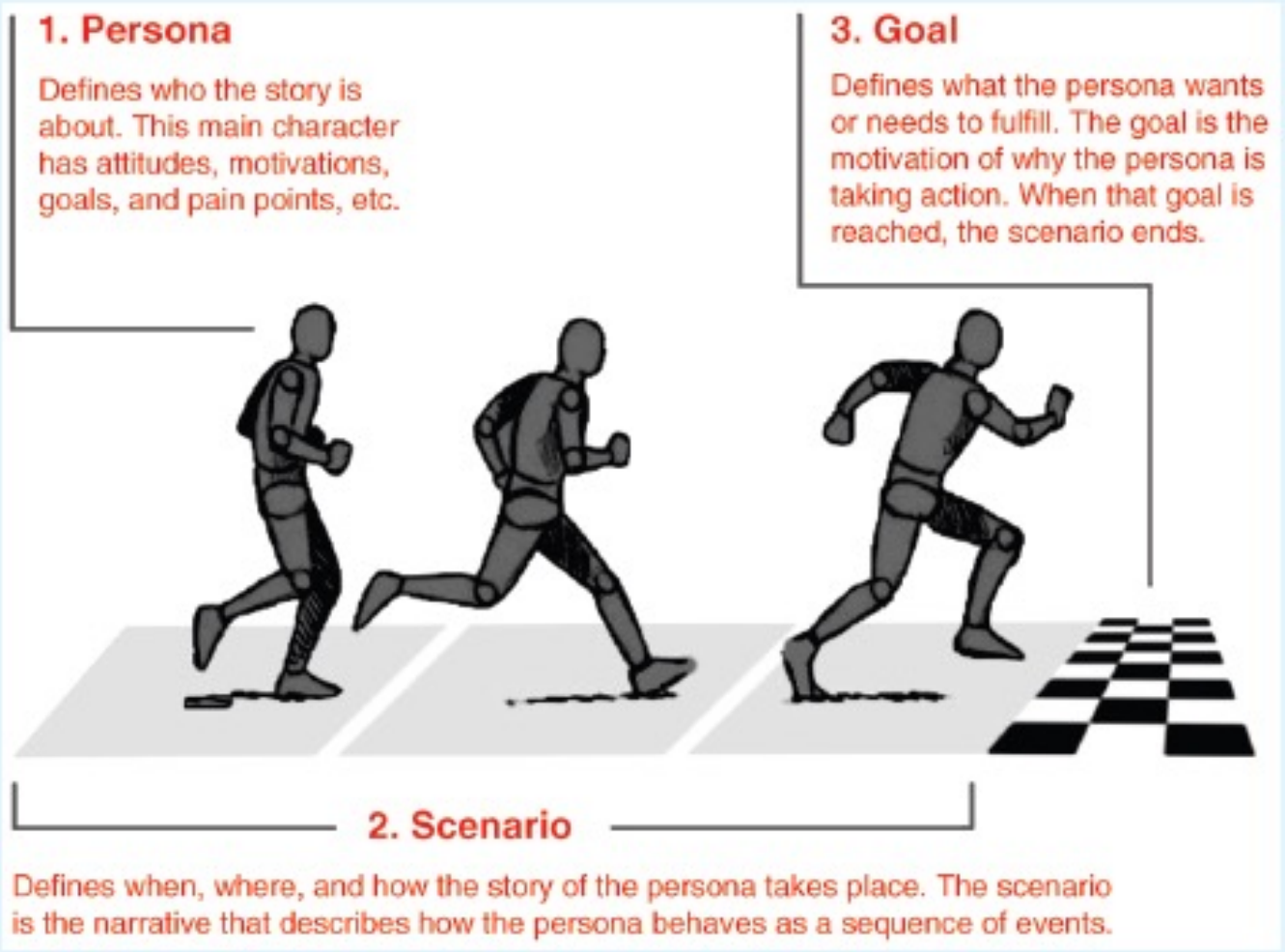


Figure 10.10 The relationship between a scenario and its associated persona

Source: http://www.smashingmagazine.com/2014/08/06/a-closer-look-at-personas-part-1/

Use cases

- Scenarios can be broken down into use cases that describe the flow of tasks that a user takes in a given functionality or path
- Use cases **focus on functional requirements** and **capture interaction**
- Use cases are **step-by-step descriptions** of interactions
- A use case involves an actor and the flow that a particular actor takes in a given functionality or path
- Two styles:
  1. **Essential use cases**: division of tasks, no implementation detail
     - In the form of [user intention, system responsibility]
  2. **Use cases with normal and alternative courses**

Example of essential use case for travel organizer

Task: Retrieve Visa

USER INTENTION	SYSTEM RESPONSIBILITY
find visa requirements	
	request destination and nationality
supply required information	
	obtain appropriate visa info
obtain copy of visa info	
	offer info in different formats
choose suitable format	
	provide info in chosen format

Note that the user intention and system responsibility are offset vertically, showing a sequence of interactions

Example of use case with normal courses for travel organizer

1. The product asks for the name of the destination country.

2. The user provides the country's name.

3. The product checks that the country is valid.

4. The product asks the user for their nationality.

5. The user provides their nationality.

6. The product checks the visa requirements of that country for a passport holder of the user's nationality.

7. The product provides the visa requirements.

8. The product asks whether the user wants to share the visa requirements on social media.

9. The user provides appropriate social media information.

Alternative courses for travel organizer

Some alternative courses:

4. If the country name is invalid:

4.1 The product provides an error message.

4.2 The product returns to step 1.

6. If the nationality is invalid:

6.1 The product provides an error message.

6.2 The product returns to step 4.

7. If no information about visa requirements is found:

7.1 The product provides a suitable message.

7.2 The product returns to step 1.



PERSONA 1 - SCENARIO 1

SCENARIOS

UX Knowledge Base Sketch #16



PERSONA 2 - SCENARIO 1

SCENARIOS ARE

- SHORT STORIES ⇒ SEQUENCE OF ACTIONS AND EVENTS FROM THE MAIN CHARACTER'S (PERSONA) PERSPECTIVE.
- ABOUT A PERSONA ⇒ BASED ON THE PERSONA'S NEEDS, GOALS, MOTIVATIONS
- ← TRYING TO ACHIEVE HER GOAL ⇒ BEFORE IMPLEMENTING SG, ASK: WOULD THIS BE USEFUL TO OUR USERS?
- BY USING THE FUTURE PRODUCT / SERVICE ⇒ IT IS ABOUT THE FUTURE!
- IN HER CONTEXT. ⇒ REAL WORLD CONTEXT CONTAINING INFORMATION ABOUT BEHAVIOR, USAGE, CONDITIONS AFFECTING THE EXPERIENCE.

SCENARIO ≠ USE CASE

- LIKE A TECHNICAL BLUEPRINT
- IT IS ABOUT FUNCTIONS, NOT THINKING OR FEELINGS
- LIST OF ACTIONS BETWEEN THE USER AND THE SYSTEM

SCENARIO ≠ USER STORY

- SHORTER (SOME SENTENCES)
- DESCRIBING A NEED FROM A PERSONA'S PERSPECTIVE
- IN THE PERSONA'S OWN WORDS.

HOW TO CREATE SCENARIOS

- PREREQUISITE: PERSONAS
- S1 S2 S3 S4 S5 — DECIDE WHAT SCENARIOS SHOULD BE CREATED: MOST SIGNIFICANT ACTIVITIES BY EACH PERSONA
- MAKE A BELIEVABLE STORY
  - ↳ PLOT: TRIGGER, ACTIONS, ACHIEVED GOAL
  - ↳ CONTEXT, E.G.: WHERE? WEATHER CONDITIONS LIGHT CONDITIONS WHEN?
  - ↳ MOTIVATIONS, REASONS
  - ↳ HOW THE FUTURE PRODUCT/SERVICE CAN HELP REACHING THE END-GOAL?
- IT IS ABOUT THE PERSONA'S EXPERIENCE (THINKING, FEELING, SEEING, DOING), NOT ABOUT UI SOLUTIONS!
- TRY TO AVOID UNREALISTIC ASSUMPTIONS! THINK ABOUT WHAT THE PERSONA WOULD DO!
- EXCLUDE THE UNNECESSARY SENTENCES!

⇒ THEN YOU CAN START DEFINING REQUIREMENTS FROM THE SCENARIOS!

Brainstorming

- Requirements may emerge directly from the data gathered, but they may also involve an innovative, creative step
- Brainstorming is a generic technique used to generate, refine, and develop ideas
- It is widely used in interaction design specifically for generating alternative designs or for suggesting new and better ideas to support users





Brainstorming: a few suggestions

- Include participants from a wide range of disciplines with a broad range of experience
- Have clear idea about the goal of the device to be designed
- First work individually and write your ideas on post-it
- Then compare each other proposals and cluster them for affinity
- **No ideas are silly or inappropriate, be open and inclusive**
  - **Don't ban silly stuff. Wild ideas often turn into really useful requirements**
- Finally, select the ideas you would like to become features of your concept design

After the requirements are discovered: the conceptual design phase

- Focuses on the idea of a product: it involves developing a conceptual model that captures what the product will do and how it will behave



- It is something that you want to test with the people you are designing for, and it's starting to look like an answer to your design challenge

- This is the moment where you move from problem to solution

Exam: the report

- The report must be maximum 12 pages long (excluding the index and references), fontsize 11, margins 2.5 cm
- The report can be written in Italian or in English
- You can attach some appendices (e.g., the presentation about the benchmarking, the design library, all the personas and scenarios, etc.), **but in the 12-page report you need to have all the relevant information** (including the most important personas and related scenarios)
- You are primarily judged on the report content, secondarily on all the complementary material you generate

Report structure

1. Executive summary (max 1 page): a clear, concise description of what you did, including methodology and results achieved
   - Here is where you may add the **motivations** for the project
2. Analysis of existing products and ideas generation
   - Examples found during the benchmarking and critical analysis of why they are relevant for your work
   - Design space identified
3. PACT analysis
   - Data gathering technique/s (procedure, results, and derive the requirements list)
   - Rich description of requirements with implications for design

Report structure

4. 1st Iteration
   - Design methodology (personas, envisioning scenarios, etc.)
   - Prototyping (low-fidelity)
   - Evaluation

5. 2nd Iteration
   - Design (adjustments done on the basis of the previous evaluation)
   - Prototyping (medium-fidelity)
   - Evaluation

5. Conclusions
   - What are the lessons learned, strengths and limitations, and possible future extensions of the project (what remains to be done? What would you do if you had more time?)

Report structure

- A mix of text explanations and pictures (photos, drawings, screenshots)
- Remember to tell about the process and what motivate your design choices, not just the final solution
- So, document everything you do, possibly when you do it (pictures, decisions, etc.): you need to demonstrate that your claims are supported by data
- Be careful with the layout and grammar
- Remember the criteria of evaluation described in the syllabus (and also reported in the slides of Lesson 1)

Homework

- Develop a primary and a secondary persona for the system of your project, based on the user research you conducted
- Develop scenarios for those personas
  - For the exam you can choose either a graphical set of vignettes, a textual description or a video

Bibliography

- Chapter 11 of the book "Interaction Design: beyond human-computer interaction" - 5th Edition by Sharp, Rogers, and Preece
- https://neohelden.com/blog/conversational-interface/apple-knowledge-navigator/